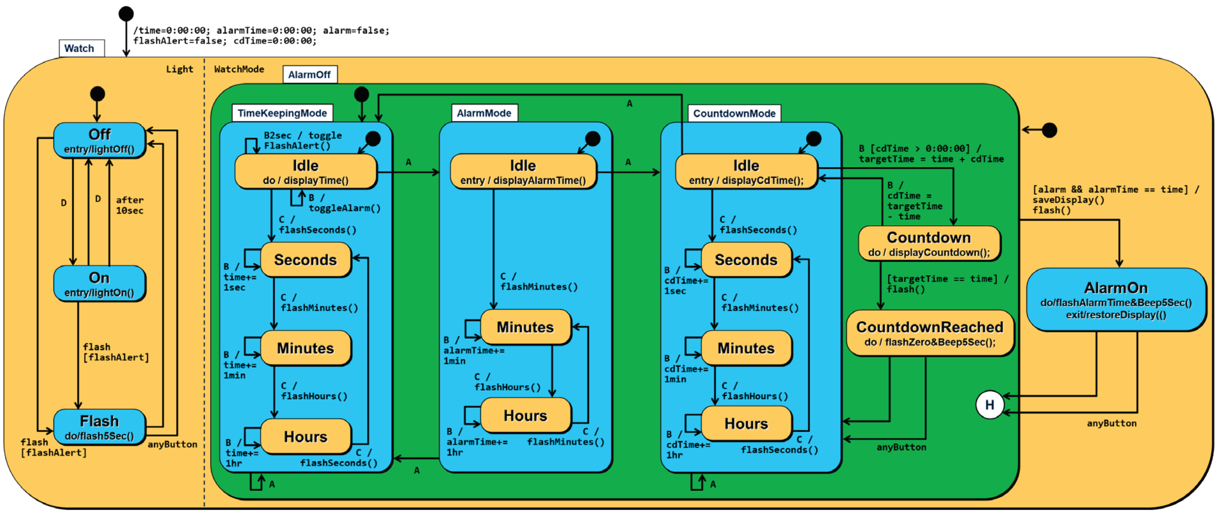
State Machine Question (100 marks)

Given the problem description below, specify the ***state machine*** for the watch. **Show the states as well as transitions including events, guards, and actions. Define composite states and orthogonal regions when needed to reduce complexity.** You do **not** need to specify parameters for events, guards, and actions **but** names of actions must clearly show when variables are initialized (and to which value) or updated (and to which value) as well as all changes to the watch display. You may assume that changing a variable will automatically update the watch display. Use **meaningful** names for the private methods for guards and actions in your state machine. You do **not** need to define the method bodies of those methods.



Alternatives: The Off and On states of Light could be placed into a composite state, so that only one flash [flashAlert] transition is needed. Instead of the flash5Sec() do action in Flash and the completion transition, it is also possible specify the startFlashing() entry action and the endFlashing() exit action and an [after 5sec] transition. The same applies to the do action of AlarmOn.

Note: The countdown is modeled with targetTime and cdTime so that the countdown can continue after being interrupted by an alarm.

Note: The scenario where the countdown is reached during an alarm is out of scope. It would require an additional transition from AlarmOn to a new state AlarmOnCountdownReached if targetTime equals time. A completion transition and an anyButton transition from this new state to Idle in CountdownMode would also be needed.